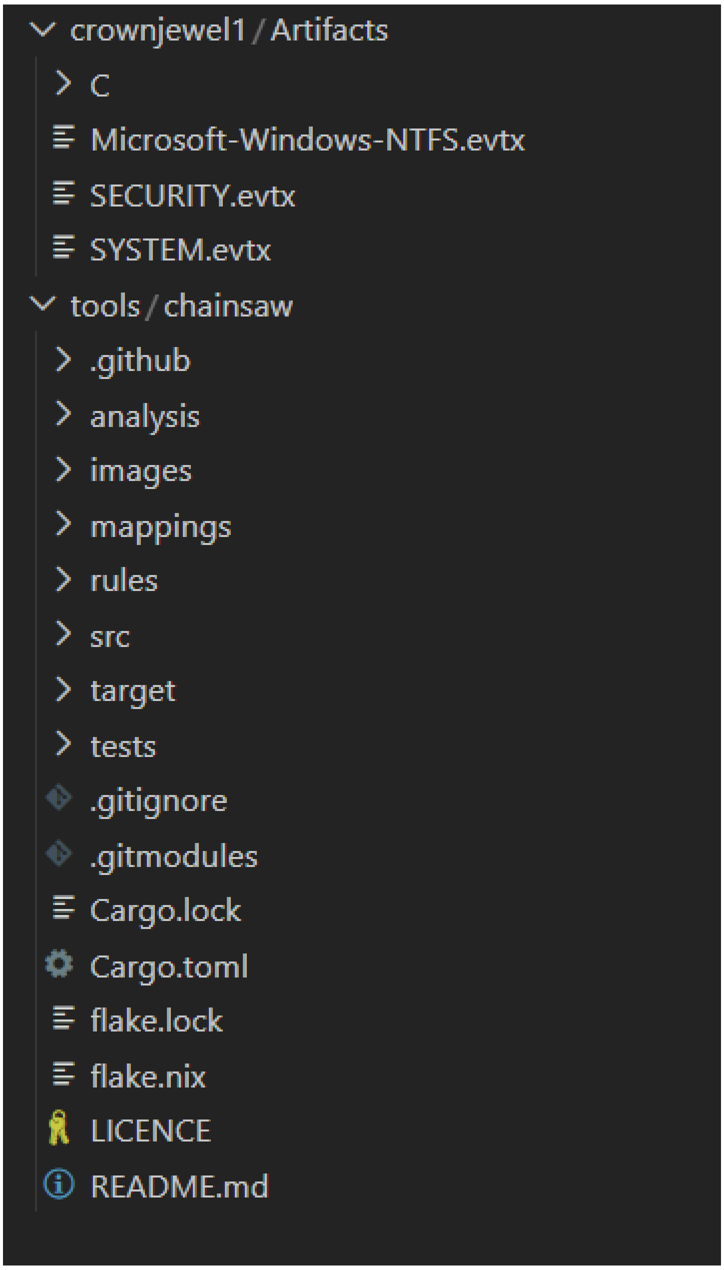
crownjewel-1 HTB



Used chainsaw to parse Windows Event Logs:

1. **~/htb_sherlock/tools/chainsaw/target/release/chainsaw search 6264 ~/htb_sherlock/crownjewel1/Artifacts/*.evtx**
2. **~/htb_sherlock/tools/chainsaw/target/release/chainsaw dump ./*.evtx --json > ./events.json**
3. **cat ./events.json | jq .[] -c**
   command used to unpack the list of events

4. **cat ./events.json | jq '.[].Event | select(.System.Channel == "System")'**
   gets every System Event log
5. **cat ./events.json | jq '.[].Event | select(.System.Channel == "System") | .System.EventID' | sort -u**
   gets the event ids for System logs

```
samba@DESKTOP-T73E2RO:~/htb_sherlock/crownjewel1/Artifacts$ cat ./events.json | jq '.[].Event | select(.System.Channel == "System") | .System.EventID'
7036
12
153
20
238
25
27
```

6. **cat ./events.json | jq '.[].Event | select(.System.Channel == "System") | .System.EventID' | sort | uniq -c | awk '{print $2 " - " $1}' | sort -n**
   arranged better the eventIDs and counted them + sorted by EventID

```
samba@DESKTOP-T73E2RO:~/htb_sherlock/crownjewel1/Artifacts$ cat ./events.json | jq '.[].Event | select(.System.Channel == "System") | .Syste
t | uniq -c | awk '{print $2 " - " $1}' | sort -n
1 - 1
3 - 2
6 - 11
12 - 2
14 - 1
15 - 1
16 - 12
18 - 1
20 - 2
```

   EventID = 14 -> volume shadow copy - vssadmin)
7. **cat ./events.json | jq '.[].Event | select(.System.Channel == "Security") | .System.EventID' | sort | uniq -c | awk '{print $2 " - " $1}' | sort -n**
8. **cat ./events.json | jq '.[].Event | .System.Channel' | sort -u**
   discovered the third category of logs:

```
"Microsoft-Windows-Ntfs/Operational"
"Security"
"System"
```

9. **cat ./events.json | jq '.[].Event | select(.System.Channel == "Microsoft-Windows-Ntfs/Operational") | .System.EventID' | sort | uniq -c | awk '{print $2 " - " $1}' | sort -n**

```
4 - 23
9 - 25
10 - 25
142 - 35
158 - 16
159 - 2
300 - 1
301 - 1
302 - 1
303 - 1
500 - 2
```

Task 1: Attackers can abuse the vssadmin utility to create volume shadow snapshots and then extract sensitive files like NTDS.dit to bypass security mechanisms. Identify the time when the Volume Shadow Copy service entered a running state.

cat ./events.json | jq '.[].Event | select(.System.EventID == 7036 and .EventData.param1 == "Volume Shadow Copy")'

7036 = a service changed state (within system evtx logs)


```
{
  "System": {
    "Provider_attributes": {
      "Name": "Service Control Manager",
      "Guid": "{555908d1-a6d7-4695-8e1e-26931d2012f4}",
      "EventSourceName": "Service Control Manager"
    },
    "EventID_attributes": {
      "Qualifiers": 16384
    },
    "EventID": 7036,
    "Version": 0,
    "Level": 4,
    "Task": 0,
    "Opcode": 0,
    "Keywords": "0x8080000000000000",
    "TimeCreated_attributes": {
      "SystemTime": "2024-05-14T03:42:16.783105Z"
    },
    "EventRecordID": 2984,
    "Correlation": null,
    "Execution_attributes": {
      "ProcessID": 784,
      "ThreadID": 916
    },
    "Channel": "System",
    "Computer": "DC01.forela.local",
    "Security": null
  },
  "EventData": {
    "param1": "Volume Shadow Copy",
    "param2": "running",
    "Binary": "5600530053002F0034000000"
  }
}
```


It’s in running state -> answer: 2024-05-14 03:42:16

Task 2: When a volume shadow snapshot is created, the Volume shadow copy service validates the privileges using the Machine account and enumerates User groups. Find the two user groups that the volume shadow copy process queries and the machine account that did it.

4799 = an account was enumerated

cat ./events.json | jq '.[].Event | select(.System.EventID == 4799)'

cat ./events.json | jq '.[].Event | select(.System.EventID == 4799) | .EventData.CallerProcessName' | sort -u

"C:\\Windows\\System32\\VSSVC.exe" - related to VSS (volume shadow copy)

"C:\\Windows\\System32\\dfsrs.exe"

"C:\\Windows\\System32\\lsass.exe"

"C:\\Windows\\System32\\svchost.exe"

cat ./events.json | jq '.[].Event | select(.System.EventID == 4799) | select(.EventData.CallerProcessName == "C:\\Windows\\System32\\VSSVC.exe") | .EventData.TargetUserName' | sort -u

```
"Administrators"
"Backup Operators"
```

answer: Administrators, Backup Operators, DC01$ (SubjectUserName)

Task 3: Identify the Process ID (in Decimal) of the volume shadow copy service process.

It's the CallerProcessID: **0x1190** **=> 4496 (int)**

```
  "EventData": {
    "TargetUserName": "Backup Operators",
    "TargetDomainName": "Builtin",
    "TargetSid": "S-1-5-32-551",
    "SubjectUserSid": "S-1-5-18",
    "SubjectUserName": "DC01$",
    "SubjectDomainName": "FORELA",
    "SubjectLogonId": "0x3e7",
    "CallerProcessId": "0x1190",
    "CallerProcessName": "C:\\Windows\\System32\\VSSVC.exe"
  }
}
```

Task 4: Find the assigned Volume ID/GUID value to the Shadow copy snapshot when it was mounted.

???????

Task 5: Identify the full path of the dumped NTDS database on disk.

cat ./mft.json | jq .[] -c | grep "ntds.dit" | jq .


```
  "StandardInfoCreated": "2024-05-14T03:44:22.813756Z",
  "FileNameFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "FileNameLastModified": "2024-05-14T03:44:22.813756Z",
  "FileNameLastAccess": "2024-05-14T03:44:22.813756Z",
  "FileNameCreated": "2024-05-14T03:44:22.813756Z",
  "FullPath": "Users/Administrator/Documents/backup_sync_dc/ntds.dit",
  "DataStreams": []
```


Task 6: When was newly dumped ntds.dit created on disk?

answer: 2024-05-14 03:44:22

Task 7: A registry hive was also dumped alongside the NTDS database. Which registry hive was dumped and what is its file size in bytes?

cat ./mft.json | jq .[] -c | grep "backup_sync_dc" | jq .


```
{
  "Signature": "FILE",
  "EntryId": 663,
  "Sequence": 4,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 1,
  "Flags": "EntryFlags(ALLOCATED)",
  "UsedEntrySize": 336,
  "TotalEntrySize": 1024,
  "FileSize": 17563648,
  "IsADirectory": false,
  "IsDeleted": false,
  "HasAlternateDataStreams": false,
  "StandardInfoFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "StandardInfoLastModified": "2023-03-25T14:40:36.021159Z",
  "StandardInfoLastAccess": "2024-05-14T03:44:42.485735Z",
  "StandardInfoCreated": "2024-05-14T03:44:42.344933Z",
  "FileNameFlags": "FileAttributeFlags(FILE_ATTRIBUTE_ARCHIVE)",
  "FileNameLastModified": "2024-05-14T03:44:42.344933Z",
  "FileNameLastAccess": "2024-05-14T03:44:42.344933Z",
  "FileNameCreated": "2024-05-14T03:44:42.344933Z",
  "FullPath": "Users/Administrator/Documents/backup_sync_dc/SYSTEM",
  "DataStreams": []
}
```


answer: SYSTEM, 17563648